

Degree in Data Science and Engineering

Lab Unit 2: Kernel Methods

Machine Learning II • J. Balsa • Academic year 2026–2027

Where are the expensive houses in California?

Practice lab. Not graded and nothing to submit.

A linear classifier can only separate classes with a hyperplane. A kernel lets it work in a richer feature space without ever building that space. In this lab we test the idea on real data: using only the location of a neighbourhood, can we tell whether its houses are expensive?

Core question. Can a kernel turn a problem that a linear classifier cannot solve into one it can?

Objectives

- See why linear classifiers fail on data that is not linearly separable.
- Compare explicit feature maps with the kernel trick.
- Understand what γ and C do in a kernel SVM.
- Match the geometry of each kernel family to the structure of a problem.
- Design a kernel for a given problem and select a model with cross-validation.

Dataset

California Housing (1990 US Census). Each row is a *block group*: a small neighbourhood of about 600–3,000 people, with its location, income, housing characteristics and median house value. The notebook downloads it from <https://raw.githubusercontent.com/ageron/handson-ml2/master/datasets/housing/housing.csv>.

Steps

1. **The problem.** Plot the data on a map and define a binary task: expensive vs. affordable neighbourhoods.
2. **Linear classifiers.** Fisher's discriminant, logistic regression and a linear SVM.
3. **Feature maps vs. kernel trick.** Hand-built polynomial features vs. the polynomial kernel.
4. **RBF kernel.** Effect of γ .
5. **Kernel families.** Linear, polynomial and RBF boundaries.
6. **Soft margin.** Effect of C on the boundary and the support vectors.
7. **Final experiment.** Add income, design your own kernel and pick the best model with cross-validation.

Questions

Question 1 — The problem

- (a) Where are the expensive neighbourhoods? What shape do they form on the map?
- (b) Is the dataset linearly separable? Is it separable by any boundary at all?

Question 2 — Linear classifiers

- (a) How different are the three decision boundaries?
- (b) Would another linear criterion or a better solver fix the problem? Relate your answer to the XOR example.

Question 3 — Feature maps vs. kernel trick

- (a) What do explicit polynomial features and the polynomial kernel have in common? What does the kernel trick save?
- (b) With $n = 3$ features and degree $d = 8$, what is $\dim \mathcal{H}$? And for the RBF kernel?
- (c) Why does the polynomial kernel barely beat the linear models here?

Question 4 — The RBF kernel

- (a) What does γ control?
- (b) For which values of γ do you see underfitting and overfitting? Justify with your results.
- (c) What happens as $\gamma \rightarrow 0$ and as $\gamma \rightarrow \infty$?

Question 5 — Kernel families

- (a) Describe the decision boundary of each kernel family.
- (b) Which family best fits the structure of this problem? Why?

Question 6 — Soft margin

- (a) How does C change the number of support vectors and the shape of the boundary?
- (b) Where do the support vectors lie on the map? Why there?
- (c) For $C = 1$, how many training points are needed to predict a new point? And to train the model?
- (d) Why does this problem have so many support vectors?

Question 7 — Final experiment Kernels C and D are defined in the notebook and use the same features.

- (a) Which kernel wins? Is the gap with the second-best model meaningful?
- (b) Why does D perform better than C?
- (c) Prove that k_D is a valid kernel for any $w \geq 0$.
- (d) What does k_D assume about the problem that kernel C does not?

Conclusion

Answer the core question in 3–5 lines. Can a kernel turn a problem that a linear classifier cannot solve into one it can? At what cost, and what did you have to decide along the way?